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Research Statement

I am a PhD candidate at the Hong Kong University of Science and Technology, and my research lies at the intersection of industrial organization and the digital economy. Within this space, my primary interest is the design of e-commerce return policy. In particular, I study how a return policy shapes purchase and return together, and whether the welfare implication changes when the return margin is ignored. Beyond returns, I am also interested in consumer search. Specifically, I study how consumers with different search costs differ in the products they search, and how this search behavior changes a seller's incentive to offer different product types. In addition, I am interested in the environmental consequences of resale price maintenance when a seller cannot lower price to clear unsold stock. I pursue the first project with Professor AU Pak Hung and the second with Professor Kohei Kawaguchi and Professor Kei Kawai. The following is a brief description of my current working papers and future research plans.

Research Papers

Return policy cannot be judged by the return rate, because the rate feeds back into price. In my job market paper, "Designing Return Policy in E-Commerce: A Trade-off between Purchase and Return Efficiency," generous reimbursement encourages purchase, but buyers do not internalize the seller's handling loss, so returns can exceed the efficient level. A stricter policy curbs those returns, yet fewer returns make a higher price more attractive and can exclude efficient purchases. In six apparel categories, a lenient rule expands trade where margins and salvage value are high, while a strict rule does better where excess returns dominate.

In "Pass-Through, Welfare, and Incidence under Product Returns in Imperfect Competition," with Ce Bian and Liang Zhao, we study pass-through and incidence with product returns. In contrast to the case without returns, a tax such as a platform commission fee or a sales tax can create negative pass-through and improve social welfare. When a high price induces consumers to return more often, the seller supplies less because the effective margin is eroded. This downward-sloping supply curve can lower the equilibrium price in response to a tax increase, raising social surplus by reducing the return rate.

Search costs shape not only whether consumers search but which products they search. In "Product Design with Heterogeneous Search Costs," with AU Pak Hung, we build on Wolinsky's (1986) sequential search model and introduce search-cost heterogeneity to study

how firms use observable product design to steer search. In equilibrium, low-search-cost consumers seek niche products and high-search-cost consumers seek standardized products, so a fall in search costs can raise both prices when the induced shift in who searches which product dominates the search-encouraging effect. Prices respond asymmetrically to where in the search-cost distribution a shock lands: technology that benefits low-cost searchers is anti-competitive, while technology that benefits high-cost searchers is pro-competitive.

Price commitment can clear the market or fill the landfill. In "Too Much Unsold Stock?: Environmental Consequences of Price Commitment," with Kohei Kawaguchi and Kei Kawai, we study resale price maintenance in the Japanese publishing industry, where publishers set retail prices and unsold books are returned upstream and discarded. Price commitment encourages inventory when demand may be low but blocks markdowns, so unsold units pile up at an environmental cost. We build a dynamic oligopoly model with demand uncertainty, estimate it on monthly bookstore data, and ask how much unsold stock is due to price commitment and how a deposit-refund tax can reduce waste.

Going forward

First, under competition the exclusion in the job market paper is less severe than under monopoly, because a consumer facing the higher price from a strict return policy can switch to other products. Second, on a two-sided platform, the same tightening of consumer returns changes seller profitability and the mix of high-value and high-return buyers. Third, Amazon's threshold-based return fees, in place since 2024, give sellers a reason to lower return rates, but whether they raise or cut price depends on whether return comes from ex-post shocks or from match value. Fourth, when products are complements or substitutes, bundling has to be judged by joint purchase and joint return together.